

Introduction to Contract Theory and Auctions

What is Contract Theory? Why study Auction Theory?

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ECON 403/503
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- **Official Prerequisites:** ECON201 and ECON301 with C- or above.
- **Unofficial Prerequisites:** Game Theory, ECON302.
- The class complements Economics of Information (ECON335) and Advanced Microeconomics (ECON305).
- A strong background in mathematics (calculus, linear algebra, probability/statistics) is recommended. Real Analysis is a plus.

- **Textbooks:**

- Maschler, M., Solan, E., & Zamir, S. (2013). *Game Theory*. Cambridge University Press.
- Milgrom (2003). *Putting Auction Theory to Work*. Cambridge University Press.
- Krishna (2009). *Auction Theory*. Academic Press.

- **Quick Game Theory References:**

- Check all the books I've posted in Moodle.
- Omer Tamuz's Caltech Notes (parts of the course are based on these).

- **Math Prerequisites:**

- Simon and Blume. *Mathematics for Economists*.
- Carter. *Foundations of Mathematical Economics*.

ECO403 (Undergraduate)	ECO503 (Master's)
Problem Sets (15%)	Problem Sets (15%)
Essay (10%)	Research Proposal (20%)
Midterm (30%)	Midterm (25%)
Final Exam (35%)	Final Exam (30%)
Post-lecture Quizzes (5%)	Post-lecture Quizzes (5%)
Midsemester Project (5%)	Midsemester Project (5%)

- Arrow-Debreu model does not apply if a trading partner has private information about some *common value* (Akerlof 1970)
- E.g., Buyer draws inferences about the quality of a car from the mere fact that the seller supplies it.
- Details of the market environment matter, and typically game theory has been used to predict market outcomes.
- This revolution has deeply changed our views on how insurance, financial, labor markets work, private transactions, etc. work (Riley 2001).

Standard Paradigm

- Exchange of goods in perfectly competitive frictionless markets
 - Perfect information
 - No externalities
 - No strategic interaction
- Rational agents take market prices as given to decide on actions
- Market prices adjust to equate demand with supply → **Market Equilibrium**
- Its theoretical apex: general equilibrium theory
 - Existence
 - Efficiency
 - First welfare theorem: Equilibrium is efficient
 - Second welfare theorem: Any efficient allocation can be achieved as an equilibrium after appropriate lump-sum transfers
- Later reformulated to account for uncertainty.
 - von Neumann and Morgenstern (1944): Preferences over uncertain outcomes
 - Arrow (1964) and Debreu (1959): State contingent commodities

There is more to market transactions

- Market: **matches** sellers and buyers.
- The match is not the end: participants make decisions/actions that affect the **value of the match**.
- Workers decide how carefully or how much effort to work.
- Buyers can choose between multiple versions of a product/multiple options for delivery.
- Sellers may restrict the supply to alter the selling price.
- Both parties' behavior depends on the **incentives** they face.

- Standard paradigm: Imperfect information is only a transaction cost
- Information economics: No!
- Small imperfections in information can have large consequences
- Why?
 - Agents' choices reveal information
- Standard paradigm: large number of agents (competitive)
- But, as soon as a small number of agents engage in a relationship and they make relationship-specific investments, they enter into a small-number bargaining situation. Does this lead to inefficiency?
- *Coase Theorem: in the absence of transaction costs the outcome of private bargaining is efficient.* → Failure of the Coase Theorem.

- During the last 50 years or so most exciting developments in economics happened in information economics.
- The traditional branch of economic theory dealing with the study and design of incentives is called contract theory.
 - Consequences of informational asymmetries between actors
 - Optimal design of contracts, institutions
- Principal-Agent(s) framework: interaction between the principal (designs the incentives) and agent(s) (usually better informed and reacts strategically.)

- The nature of the information asymmetry, and the timing of the contract design matters:
 - Hidden-Action: the agents can chose among several actions, which the principal doesn't observe.
 - Hidden-information: The agent holds private information which is relevant to the transaction.
 - Signaling: The informed agent has payoff relevant information that may choose to costly disclose.
 - Cheap-Talk: Communication between agents, or principal-agent, in which the communication does not directly affect payoffs.
 - Bayesian Persuasion/Information design: the principal commits to a communication protocol, which is designed optimally to nudge the agent to act in a certain way.

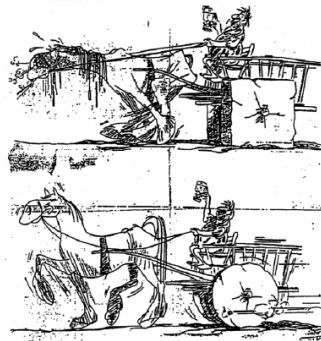
An impressive number of diverse areas including

- Design of institutions and organizations
- Corporate finance
- Regulation, auditing
- Development economics
- Macroeconomics
- Public finance: optimal taxation
- Allocation of ownership: vertical integration, privatization
- Accounting
- Law
- Political Science

- A specific kind of contract or market mechanism is an **auction**.
- 2020's Nobel in Economics was awarded to Bob Wilson and Roger Myerson for their pioneering contribution to auction theory.
- Spectrum licensing, mineral and oil rights, clean energy procurement, utilities procurement, internet advertisements, art, fresh fish.
- Why learn auction theory?
 - The fundamental building-blocks helps understand more complex environments.
 - Tools provide connections to other branches of 'standard' economic theory.
 - Develop intuition and insights that can inform other areas in economics.

Why Every Economist Should Learn Some Auction Theory

Paul Klemperer, August 2001.



Auction Theory **Heineken.**
Refreshes the parts other
economics **beers cannot reach.**

- ① The classical auction formats under complete information.
- ② Modelling information and beliefs: knowledge spaces and belief spaces.
- ③ Advanced Introduction to Game Theory: Games of Incomplete Information.
- ④ Bayesian Auctions: classical results and extensions.
- ⑤ Advanced topics, but depends on class performance/interests.